

Traces, Powers, and Closed Patterns

Adjacency matrices as relations — and how the same loop reappears in the Frobenius trace of this project

L linearized trace **F** Frobenius **C** cyclotomic cosets

A follow-up picture book, in plain English, with categories and TikZ

One idea, three costumes. The trace of a matrix adds up its diagonal; that already *counts self-relations* ($x R x$). Raise the matrix to a power first and the diagonal counts something richer — **closed walks**, i.e. patterns that return to where they started. Categorically this is nothing but *feedback*: the trace of R^k picks out the points fixed by composing the relation k times. The punchline for *this* formalisation: the Frobenius map $x \mapsto x^2$ is itself an endorelation, its “feedback” cycles are exactly the cyclotomic cosets **C**, and the project’s trace $\text{Tr}(x) = \sum_{i < n} x^{2^i}$ **L** is the very same categorical loop applied to “multiply by x ”.

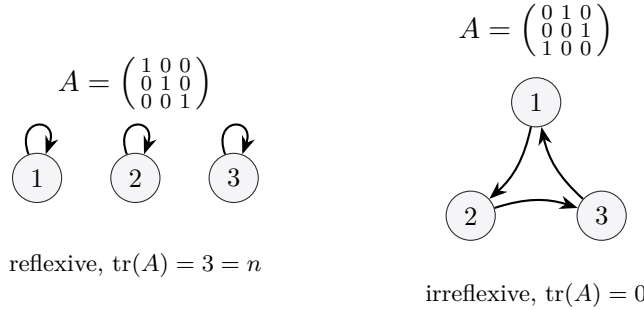
1. The trace already counts self-relations

Let R be a relation on a finite set $\{1, \dots, n\}$: write $a_{ij} = 1$ when $i R j$ and 0 otherwise. The array $A = (a_{ij})$ is the **adjacency matrix**, and a relation is literally the same data as its 0/1 matrix. Its trace is

$$\text{tr}(A) = \sum_i a_{ii} = \#\{i : i R i\} = \text{number of self-loops}.$$

So before any powers appear, the trace is already a counting statement:

$$\begin{array}{ll} \text{Reflexive } (x R x \text{ always}) & \text{tr}(A) = n \\ \text{Irreflexive } (\text{never } x R x) & \text{tr}(A) = 0 \end{array}$$



2. Powers of the matrix compose the relation

Multiplying adjacency matrices *composes* relations. The Boolean reading is “there is a step”: $(A^2)_{ij} \neq 0$ exactly when some k has $i R k R j$. The linear-algebra reading *counts the ways*:

$$(A^k)_{ij} = \#\{\text{walks } i \rightarrow \dots \rightarrow j \text{ of length } k\}.$$

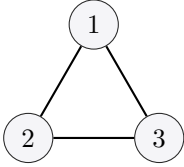
Putting $j = i$ and summing the diagonal gives the headline formula.

$$\text{tr}(A^k) = \sum_i (A^k)_{ii} = \#\{\text{closed walks of length } k\}.$$

Each closed walk starts and ends at the same vertex; equivalently it is a **fixed point of $R^k = R \circ \dots \circ R$** , i.e. a cycle whose length divides k .

$k = 2$ **detects symmetry.** Since $(A^2)_{ii} = \sum_j a_{ij}a_{ji}$, each term is 1 precisely when $i \leftrightarrow j$ is a *mutual* pair. So $\text{tr}(A^2)$ measures how symmetric the relation is; for an undirected graph every edge is counted twice, $\text{tr}(A^2) = 2 \cdot (\# \text{edges}) = \sum_i \deg(i)$.

$k = 3$ **counts triangles.** $\text{tr}(A^3)$ counts closed walks of length 3; for an undirected simple graph each triangle is walked 6 ways (3 starts $\times$ 2 directions), so $\# \text{triangles} = \text{tr}(A^3)/6$.

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$


$$A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{pmatrix}$$

undirected triangle

$$\text{tr}(A) = 0, \quad \text{tr}(A^2) = 6 = 2 \cdot 3 \text{ edges},$$

$$\text{tr}(A^3) = 6 \Rightarrow \frac{6}{6} = 1 \text{ triangle}.$$

For the *directed* 3-cycle of §1, instead $\text{tr}(A) = \text{tr}(A^2) = 0$ but $A^3 = I$, so $\text{tr}(A^3) = 3$: three vertices each return after three directed steps — one directed triangle, walked from each of its 3 starts.

3. A dictionary: relation properties $\leftrightarrow$ traces

Property	Matrix condition	Trace role
Reflexive	$a_{ii} = 1$	$\text{tr}(A) = n$
Irreflexive	$a_{ii} = 0$	$\text{tr}(A) = 0$
Symmetric	$A = A^T$	large $\text{tr}(A^2)$ (every mutual pair counts)
Antisymmetric	$a_{ij}a_{ji} = 0$ ($i \neq j$)	$\text{tr}(A^2) = \text{tr}(A)$
Transitive	$A^2 \leq A$ (Boolean product)	<i>no</i> simple trace formula

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{tr}(A^2) = \text{tr}(A) = 0$$

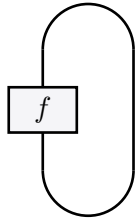


antisymmetric: no $i \leftrightarrow j$

Why transitivity is the odd one out. The first four rows are *linear-algebra* facts: they count things, and counting is what trace does. Transitivity ($A^2 \leq A$) is a *Boolean* condition — it asks only whether a length-2 path *exists*, never how many. Existence is not a sum of a diagonal, so there is no clean trace characterization. This is the honest boundary of the trace story: *trace lives over linear algebra, transitivity lives over Boolean algebra*.

4. The categorical picture: trace = feedback

Strip away the matrix and only the arrows remain. An **endorelation** $R: X \rightarrow X$ is an arrow from an object to itself; composing it is taking powers, $R^k = R \circ \dots \circ R$. In a symmetric monoidal category with duals the trace of an endomorphism $f: V \rightarrow V$ is the single closed loop (evaluation ε against coevaluation η):

$$I \xrightarrow{\eta} V \otimes V^* \xrightarrow{f \otimes \text{id}} V \otimes V^* \xrightarrow{\text{swap}} V^* \otimes V \xrightarrow{\varepsilon} I$$


$$= \text{tr}(f)$$

Applied to the adjacency matrix this loop is exactly $\sum_i a_{ii}$; applied to A^k it is $\sum_i (A^k)_{ii}$. So the three faces of the same idea are:

trace = local invariant: the closed loops / self-feedback of R ,
 R^k = the relation composed with itself k times,
 $\text{tr}(R^k)$ = the **fixed points** of R^k : points returning after k steps.

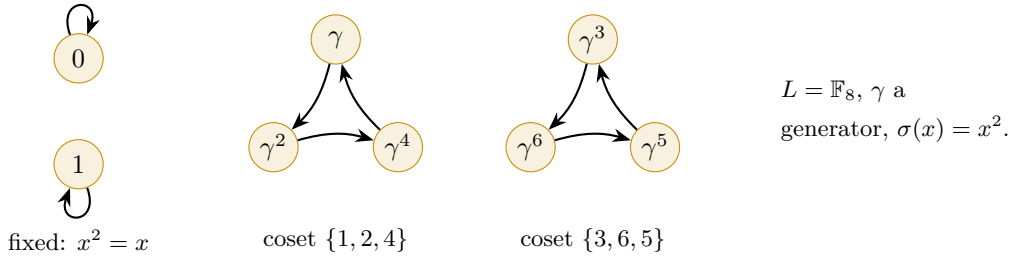
This “count the points that come back” reading is why traces of powers pervade graph theory, spectral graph theory, and topology — the **Lefschetz fixed-point theorem** is the same slogan for continuous maps: an alternating sum of traces on homology counts (with signs) the fixed points of a self-map. Endorelation, endomorphism, endomap: one loop, many settings.

5. Where this project lives: the Frobenius endorelation

Now specialise the object to a finite field $L = \mathbb{F}_{2^n}$ of characteristic 2, the setting of the [Equation1](#) formalisation. Two different “adjacency” pictures appear, and the trace loop of §4 explains both.

5a. Frobenius as a graph — its cycles are the cosets

The Frobenius map $\sigma: x \mapsto x^2$ is a *bijective* endorelation on L (brick ). A bijection’s graph is a permutation matrix P , and its picture is a disjoint union of cycles — its **functional graph**. Here those cycles are precisely the **cyclotomic cosets** (brick ): the orbit of an exponent e under doubling mod $2^n - 1$.



Reading off the trace of powers of this permutation matrix P is now pure §4: $\text{tr}(P^k)$ counts the elements fixed by σ^k , i.e. solutions of $x^{2^k} = x$, i.e. the subfield $\mathbb{F}_{2^{\text{gcd}(k,n)}} \cap L$:

$$\text{tr}(P^k) = \#\{x \in L : x^{2^k} = x\} = 2^{\text{gcd}(k,n)}.$$

For $L = \mathbb{F}_8$ ($n = 3$): $\text{tr}(P) = \text{tr}(P^2) = 2$ (just 0, 1) and $\text{tr}(P^3) = 8$ (everything returns) — the two length-3 cosets snap shut exactly when $3 \mid k$. This is the “cycles of length dividing k ” rule of §2, now saying **cyclotomic cosets** = **Frobenius cycles**, the concrete content of brick .

5b. The project’s trace is the loop of §4 — brick

The *other* matrix is linear, not Boolean. Fix $x \in L$ and multiply by it: $m_x: L \rightarrow L$, $y \mapsto x \cdot y$. Viewing L as an n -dimensional $\mathbb{F}_2$ -vector space, m_x has an honest 0/1-and-beyond matrix over $\mathbb{F}_2$, and the categorical loop of §4 applied to it is the **field trace**. Over $\mathbb{F}_2$ that loop unwinds into a Frobenius sum — the project’s own $\text{Tr}/\text{truncTrace}$ (brick ):

$$\text{Tr}_{L/\mathbb{F}_2}(x) = \text{tr}(m_x) = \sum_{i=0}^{n-1} x^{2^i}.$$

Machine-checked in Equation1/TraceLibrary.lean:

`Tr_eq_truncTrace` — the two trace functions coincide;

`Tr_eq_algebraMap_trace` — $\text{Tr } nx = \text{algebraMap } \mathbb{Z}/2 \ L (\text{Algebra.trace } \mathbb{Z}/2 \ L \ x);$

so the diagonal-sum trace of m_x is $\sum_{i < n} x^{2^i}$ (via `FiniteField.algebraMap_trace_eq_sum_pow`).

Notice the pleasing overlap between the two pictures. The summands $x, x^2, x^4, \dots$ of the trace are the images of x under the Frobenius orbit of §5a — **summing an element around its own coset cycle**. The feedback loop of §4 (add up the diagonal of m_x) and the fixed-point count of §5a (trace of σ^k) are two readings of the one word “trace”: the first over linear algebra **L**, the second over the Boolean/permutation combinatorics of the cosets **C**, glued by the Frobenius **F**.

6. Does it apply to Dobbertin? — yes, with one caveat

Applicable. The trace $\text{Tr}(x) = \sum_{i < n} x^{2^i}$ that Dobbertin uses to state equation (1) is exactly the categorical trace of “multiply by x ” (§5b, verified in `TraceLibrary.lean`), and the doubling/cyclotomic-coset bookkeeping he uses silently is the Frobenius functional graph of §5a.

Caveat. The trace of powers identity $\text{tr}(A^k) = \#\text{closed walks}$ is a statement about the 0/1 adjacency matrix of a general relation; in the Dobbertin/Equation1 extract it appears in the specialised *permutation* form of §5a ($\text{tr}(\sigma^k) = 2^{\text{gcd}(k,n)}$), not as a graph-triangle count. The transitivity row of §3 has, as noted there, no trace avatar — and correspondingly no role in this field-trace development.